NAME : .....

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) — path to success

RACE
Regular Analysis through Continuous Exercise

JEE (Main + Advanced) 2023
ENTHUSIAST + LEADER COURSE

SCORE ADVANCED | **MATHEMATICS** | **M.M. 62**

REVISION ASSINGMENT # 04 | (P & C, BT) | TIME : 60 MIN.

SECTION–I

Straight Objective Type (3 Marks each, –1 for wrong answer)

1. If the domain and range of $f(x) = {}^{9-x}C_{x-1}$ contains m and n elements respectively, then -

(A) m = n (B) m = n + 1 (C) m = n – 1 (D) m = n + 2

2. If n balls of identical weight are divided into groups of three balls each, then and sum of the weights of all possible groups is 480 gms. When they are weighed in group of four balls the sum of the weights come out to be 640 gms for all possible groups, then n is :-

(A) 7 (B) 6 (C) 8 (4) None of these

3. Number of ordered triplets (a, b, c) of positive integers less than 10, for which the product abc is divisible by 20, are -

(A) 48 (B) 102 (C) 54 (D) 51

4. The coefficient of x^4 in the expansion $(e^x - 1)^2$ is :

(A) $\frac{3}{8}$ (B) $\frac{5}{12}$ (C) $\frac{7}{12}$ (D) $\frac{5}{24}$

5. The sum of the series $(1^2+1)1! + (2^2+1)2! + (3^2+1)3! + ... + (n^2+1)n!$ is :

(A) $(n+1)((n+2)!)$ (B) $n(n+1)!$ (C) $(n+1)(n+1)!$ (D) none of these

6. The value of ${}^{3n}C_n - {}^nC_1 \cdot {}^{3n-3}C_n + {}^nC_2 \cdot {}^{3n-6}C_n - {}^nC_3 \cdot {}^{3n-9}C_n + ...$ is equal to :-

(A) 1 (B) 2^n (C) 3^n (D) $3^n - 2^n$

7. $\sum\limits_{0\le i<j\le n} \sum (i+j)(C_i^2 + C_j^2 + C_iC_j)$ is equal to :-

(A) $\frac{n}{2}\{(2n-1)\,{}^{2n}C_n + 4^n\}$ (B) $\frac{n}{2}\{(2n+1).\,{}^{2n}C_n + 4^n\}$

(C) $\frac{n}{2}\{4^n - (2n-1).{}^{2n}C_n\}$ (D) $\frac{n}{2}\{4^n - (2n+1).\,{}^{2n}C_n\}$

Multiple Correct Answer Type (4 Marks each, –1 for wrong answer)

8. If $\sum\limits_{r=0}^{n} \frac{r}{{}^nC_r} = \sum\limits_{r=0}^{n} \frac{n^2 - 4n + 6}{2.{}^nC_r}$, then -

(A) n = 1 (B) n = 2 (C) n = 3 (D) n = 4

9. If set A contains m elements, B and C are two subsets of A such that n(B) = n(C) ; $B \cup C = A$; $n(B \cap C) = 1$

then identify the correct statements :-

(A) m must be odd

(B) m must be even

(C) Number of ways in which subset B and C are to be formed is $m.\,{}^{m-1}C_{\frac{m-1}{2}}$

(D) Number of ways in which subsets B and C are to be formed is $\frac{1}{2}.\,{}^{m-1}C_{\frac{m-1}{2}}$

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10. If the expansion of $(2x - 3y + z)^{10} = \sum_{\substack{0\le\alpha,\beta,\gamma\le 10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!}(2x)^{\alpha}(-3y)^{\beta}(z)^{\gamma}$, then -

(A) the expansion will contain 66 terms

(B) $\sum_{\substack{0\le\alpha,\beta,\gamma\le 10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!} = 3^{10}$

(C) the coefficient of $x^2y^3z^5$ will be $\left(-\frac{9\cdot 10!}{5!}\right)$

(D) $\sum_{\substack{0\le\alpha,\beta,\gamma\le 10\\ \alpha+\beta+\gamma=10}} \frac{10!}{\alpha!\beta!\gamma!}2^{\alpha}(-3)^{\beta} = 0$

11. The number of selections of fruits from 4 mangoes, 3 oranges, 2 papayas and 2 pineapples will be [If fruits of same type are treated as same] -

(A) 179 (when atleast one fruit is chosen)

(B) 175 (atleast 2 fruits are chosen)

(C) 15 (at most 2 fruits are chosen with atleast one fruit)

(D) 15 (none or more fruits but not more than 2 fruits are chosen)

Linked Comprehension Type (Single Correct Answer Type) (3 Marks each, –1 for wrong answer)

Paragraph for Question 12 to 14

If a man climbs either one step or k steps at a time then total number of ways in which he can climb up a staircase of 3k steps (starting from the bottom) where $k \ge 2$ is denoted as A(1, k)

12. A(1, 2) is equal to :-

(A) 10 (B) 11 (C) 12 (D) 13

13. If A(1, k) = 26 then k is equal to :-

(A) 2 (B) 3 (C) 4 (D) k

14. If $A(1, k+1) - A(1, k) = \lambda_k$ then $\lambda_{k+1} - \lambda_k$ is equal to :-

(A) 1 (B) 2 (C) 0 (D) 4

SECTION–II

Numerical Grid Type (Upto Second Decimal place) (4 Marks each, –1 for wrong answer)

1. The remainder when $\sum_{r=1}^{10}\left({}^{10}C_r\right)(r-5)^3$ is divided by 10, is

2. If $n \in N$, then the remainder when $(57)^{n+3} + (79)^{n+1} + (46)^n$ is divided by 11 is -

3. A gentleman gives a dinner party to 5 guests to be selected from 10 friends. The number of ways of forming a party of 5, given that two particular friends will not attend party together -

4. If the sum of all the odd divisors of 10!, which are of the form 3n + 1 is S_1 and which are of the form 3n – 1 is S_2. Then the remainder obtained by dividing S_1 by S_2 is

REVISION ASSINGMENT # 03 (COMPLEX NUMBER) MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	D	B	D	D	D	C	A,B,C,D	B,D
SECTION-II	Q.	11	12	13	14						
	A.	B,C	C	B	C						
	Q.	1	2	3	4						
	A.	4.50	1.00	5.00	100.00						

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